Class 6 Science – Chapter 4: Exploring Magnets

Complete Question–Answer Notes | Simple English | CBSE

These notes are based on your uploaded 9-page PDF, Exploring Magnets. They include the textbook's "Think and Answer," "Identify and Answer," "Quick Check," and activity questions, followed by important questions from every topic. The diagrams on pages 3–6 are also covered in the explanations.

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Part 1: All textbook questions and answers

A. Think and Answer — Page 1

Q1. What are the things attached to the refrigerator door called?

Answer: They are called refrigerator magnets or magnetic fridge magnets.

Q2. Which part of these things helps them stick to the fridge?

Answer: A small magnet attached to the back helps them stick to the refrigerator door.

Q3. Do you think these things will stick to an iron almirah?

Answer: Yes, they will stick to an iron almirah because iron is attracted by a magnet.

Q4. Do you think these things will stick to a wooden table?

Answer: No, they will not stick to a wooden table because wood is a non-magnetic material.

B. Think and Answer — Page 5

Q5. Justify whether the statement is true or false: All the parts of a magnet attract all magnetic materials equally.

Answer: The statement is false. A magnet is usually strongest at its two poles. Magnetic materials are attracted more strongly near the poles than near the middle of a bar magnet.

C. Identify and Answer — Page 6

The diagram shows two bar magnets and the way their poles interact.

Q6. Which property of magnets is being shown in the diagram?

Answer: The diagram shows that like poles repel each other and unlike poles attract each other.

Q7. How did you identify this property?

Answer: The arrows show the magnets moving towards or away from each other. When unlike poles face each other, they attract. When like poles face each other, they repel.

Q8. What would happen if magnet A is broken into two pieces?

Answer: Each piece will become a smaller magnet with its own north and south poles. A single north pole or south pole cannot be obtained by simply breaking a magnet.

D. Quick Check — Page 6

State whether each statement is true or false.

No.	Statement	Answer
1	A magnet can attract objects made of wood and plastic.	False
2	A magnet has two poles.	True
3	The poles of a magnet can exist separately.	False
4	A stainless-steel spoon is attracted by a magnet.	False, in the usual textbook example
5	Magnets are used for finding directions.	True

Note: Stainless steel comes in different types. Some types are magnetic, while others are weakly magnetic or non-magnetic.

E. Do and Learn — Page 7

Q9. Can the process of repeatedly rubbing a bar magnet over an iron piece be called magnetic induction?

Answer: No. This process is called magnetisation by stroking. Magnetic induction occurs when a magnetic material becomes magnetised due to the influence of a nearby magnet without rubbing it.

Q10. What do you observe when an unmagnetised iron object is brought near the magnetised iron piece?

Answer: The unmagnetised iron object may be attracted to the magnetised iron piece because it behaves like a magnet and can induce magnetism in nearby iron.

Q11. How will you determine the poles of this iron piece?

Answer: Bring a known north pole of a bar magnet near each end of the iron piece. The end that repels the north pole is the north pole; the end that attracts it is the south pole. Repulsion is the more reliable test.

Q12. Leave the iron piece undisturbed for four days. Check again if it is still magnetised. What do you observe?

Answer: The iron piece may lose some or most of its magnetism because it is a temporary magnet. The exact result depends on the iron and how it was magnetised.

Part 2: Important questions and answers from every topic

Topic 1: What is a magnet?

Q13. What is a magnet?

Answer: A magnet is an object that attracts certain materials, such as iron, nickel and cobalt.

Q14. What is magnetism?

Answer: Magnetism is the property by which a magnet attracts certain materials or exerts forces on other magnets.

Q15. What is the origin of the word “magnet”?

Answer: The word is believed to have originated from the Greek word magnes lithos, meaning a lodestone or leading stone.

Q16. Why are magnets useful in our daily life?

Answer: Magnets are used in refrigerator doors, speakers, electric motors, compasses and many other devices.

Topic 2: Discovery of magnets

Q17. How were magnets discovered according to the traditional story?

Answer: According to a traditional story, a shepherd named Magnes was tending his sheep in Magnesia, Greece. The iron tip of his stick and the nails in his shoes were attracted to a natural magnetic rock. This rock was called a lodestone.

Q18. What is a lodestone?

Answer: A lodestone is a naturally occurring magnetic rock that can attract iron.

Q19. What is the use of a lodestone?

Answer: A freely suspended lodestone can align approximately in the north–south direction. It was used to find directions.

Q20. How are artificial magnets made?

Answer: Artificial magnets are made by magnetising suitable magnetic materials, such as iron or steel, using appropriate methods.

Q21. Name different shapes of artificial magnets.

Answer: Common shapes include:

- Bar magnet
- Cylindrical magnet
- Horseshoe-shaped magnet
- U-shaped magnet
- Dumbbell-shaped magnet

The illustrations on page 2 show these different shapes.

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Topic 3: Magnetic and non-magnetic materials

Q22. What are magnetic materials?

Answer: Magnetic materials are materials that are attracted by a magnet. Iron, nickel and cobalt are common examples.

Q23. What are non-magnetic materials?

Answer: Non-magnetic materials are materials that are not attracted appreciably by an ordinary magnet. Wood, paper, rubber and plastic are common examples.

Q24. Give five examples of magnetic materials.

Answer: Iron, nickel, cobalt, many types of steel and some iron-containing alloys.

Q25. Give five examples of non-magnetic materials.

Answer: Wood, paper, rubber, glass and plastic.

Q26. Is every metal attracted by a magnet?

Answer: No. Metals such as aluminium and copper are not attracted appreciably by an ordinary bar magnet. Iron, nickel and cobalt are attracted by magnets.

Q27. Is stainless steel always non-magnetic?

Answer: No. Some types of stainless steel are magnetic, while others are weakly magnetic or non-magnetic.

Q28. How can we separate iron pieces from waste materials?

Answer: We can use a magnet to attract and separate iron pieces from other waste materials.

Topic 4: Poles of a magnet

Q29. What are the poles of a magnet?

Answer: The poles are the two regions of a magnet where its magnetic strength is greatest.

Q30. Where is the magnetic force strongest in a bar magnet?

Answer: The magnetic force is strongest near the two ends of a bar magnet, called its north and south poles.

Q31. How can we show that a magnet is strongest at its poles?

Answer: Spread iron filings on a sheet of paper and move a bar magnet underneath it. Most iron filings collect near the two ends of the magnet, showing that its magnetic strength is greatest at the poles.

Q32. Does the middle of a bar magnet have no magnetism?

Answer: No. The middle of a bar magnet also has a magnetic field, but the magnetic strength is usually greatest near the poles.

Topic 5: Directional property of magnets

Q33. What happens when a bar magnet is suspended freely?

Answer: It comes to rest approximately in the north–south direction.

Q34. What is the north-seeking pole?

Answer: The pole of a freely suspended magnet that points towards geographic north is called the north-seeking pole or north pole.

Q35. What is the south-seeking pole?

Answer: The pole that points towards geographic south is called the south-seeking pole or south pole.

Q36. Why does a freely suspended magnet point in the north–south direction?

Answer: Earth has a magnetic field. This field exerts a force on the magnet and makes it align approximately in the north–south direction.

Q37. How does a freely suspended magnet help us find directions?

Answer: When the magnet comes to rest, one end points approximately north and the other points approximately south. We can then identify east and west.

Topic 6: Poles always exist in pairs

Q38. What happens when a bar magnet is broken into two pieces?

Answer: Each piece becomes a smaller magnet with its own north and south poles.

Q39. Can we obtain only the north pole or only the south pole by breaking a magnet?

Answer: No. Each resulting piece has both a north pole and a south pole.

Q40. What does the statement “magnetic poles always exist in pairs” mean?

Answer: It means that an ordinary magnet always has both north and south poles. We cannot obtain a single isolated pole simply by breaking the magnet.

The diagrams on page 4 illustrate how new poles appear at the broken ends of a magnet.

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Topic 7: Like poles repel and unlike poles attract

Q41. What happens when the north pole of one magnet is brought near the north pole of another magnet?

Answer: They repel each other.

Q42. What happens when the south pole of one magnet is brought near the south pole of another magnet?

Answer: They repel each other.

Q43. What happens when the north pole of one magnet is brought near the south pole of another magnet?

Answer: They attract each other.

Q44. State the law of magnetic poles.

Answer: Like magnetic poles repel each other, while unlike magnetic poles attract each other.

Q45. How can we identify the poles of an unknown magnet using a known magnet?

Answer: Bring a known pole near each end of the unknown magnet. If an end repels the known north pole, it is a north pole. If it attracts the known north pole, it is a south pole.

Topic 8: Making a magnet

Q46. What is magnetisation?

Answer: Magnetisation is the process of making a suitable material magnetic.

Q47. What is magnetic induction?

Answer: Magnetic induction is the process in which a magnetic material becomes magnetised due to the influence of a nearby magnet.

Q48. How can we make a magnet by stroking?

Answer: Place an iron object on a table. Stroke it repeatedly in the same direction with one pole of a bar magnet. Lift the magnet after each stroke and repeat the process many times.

Q49. Can an iron nail be made into a magnet?

Answer: Yes. An iron nail can be magnetised by stroking it repeatedly with a magnet.

Q50. What are temporary magnets?

Answer: Temporary magnets are magnets that lose most or all of their magnetism relatively easily when the magnetising influence is removed.

Q51. Give examples of objects that can be made into temporary magnets.

Answer: An iron nail, iron pin and iron needle can be magnetised temporarily.

Q52. What is the difference between magnetic induction and stroking?

Answer: In magnetic induction, a material becomes magnetised due to the influence of a nearby magnet without rubbing. In stroking, a magnet is repeatedly moved along the material in one direction to magnetise it.

Topic 9: Magnetic compass

Q53. What is a magnetic compass?

Answer: A magnetic compass is an instrument used to find directions. It contains a magnetised needle that can rotate freely.

Q54. Who used an early direction-finding device described in the chapter?

Answer: The chapter describes a traditional story about the Chinese emperor Huang Ti, whose direction-seeking chariot used a rotating statue.

Q55. How does a magnetic compass work?

Answer: The compass needle aligns with Earth's magnetic field and comes to rest approximately in the north–south direction.

Q56. What is the purpose of the markings on a compass dial?

Answer: The markings show directions such as north, south, east and west, helping us determine our direction.

Q57. How can we use a compass to find directions?

Answer:

1. Place the compass on a level, stable surface.
2. Keep it away from magnets and other objects that may affect the needle.
3. Wait for the needle to stop moving.
4. Identify the north end of the needle.
5. Use the compass dial to identify the other directions.

Topic 10: Making a magnetic compass

Q58. How can we make a simple magnetic compass?

Answer:

1. Magnetise an iron needle by stroking it repeatedly with a bar magnet in one direction.
2. Place the needle on a small piece of cork or foam.
3. Float it in a bowl of water.
4. Allow the needle to come to rest.

5. Observe that it points approximately in the north–south direction.

Q59. Why is cork or foam used in a simple compass?

Answer: Cork or foam floats on water and allows the magnetised needle to rotate freely.

Q60. Why must the needle not touch the water?

Answer: The needle must rest on the floating cork or foam so that it can move freely without being hindered by the water.

Topic 11: Keeping magnets safe

Q61. How can a magnet lose its magnetic properties?

Answer: A magnet may lose some of its magnetism if it is heated strongly, hammered, dropped repeatedly or stored improperly.

Q62. How should bar magnets be stored?

Answer: Bar magnets should be stored in pairs with unlike poles next to each other and a wooden spacer between them, with soft iron keepers across the ends as appropriate for the storage arrangement.

Q63. What is a magnetic keeper?

Answer: A magnetic keeper is a piece of soft iron placed across the poles of a magnet to help preserve its magnetism during storage.

Q64. How should a horseshoe magnet be stored?

Answer: A soft iron keeper should be placed across its two poles.

Q65. Why should magnets be kept away from some electronic devices?

Answer: Strong magnets can interfere with magnetic sensors, speakers, storage devices and other sensitive electronic components. They should be kept away from devices that may be affected.

The safe-storage illustrations on pages 8–9 show bar magnets stored with a spacer and a horseshoe magnet stored with a keeper.

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Topic 12: Uses of magnets

Q66. Write five uses of magnets.

Answer:

1. Magnets help refrigerator doors remain closed.
2. Magnetic compasses help us find directions.
3. Magnets are used in speakers, headphones and microphones.
4. Magnets are used in electric motors.
5. Magnets help separate iron from scrap materials.

Q67. How are magnets used in ATM cards and credit cards?

Answer: Traditional magnetic-stripe cards store information in a magnetic strip.

Q68. How are magnets used in speakers?

Answer: Magnets interact with electric currents in a speaker to produce vibrations that create sound.

Q69. How are magnets used in electric motors?

Answer: Magnets create magnetic fields that interact with electric currents to produce motion in electric motors.

Q70. How are magnets useful in scrapyards?

Answer: Powerful magnets can lift and separate iron and steel objects from mixed scrap materials.

Q71. Why are magnets used in refrigerator doors?

Answer: Magnets help keep the door tightly closed and prevent it from opening easily.

Part 3: Activity-based questions and answers

These answers cover the practical activities shown on pages 2–8 of your PDF.

Q72. What is the aim of the activity on page 2?

Answer: The aim is to show that magnets attract iron objects.

Q73. What is observed when a bar magnet is rolled over a mixture of iron pins, paper, rubber bands, cloth pieces and pencil shavings?

Answer: The iron pins stick to the magnet, while paper, rubber bands, cloth pieces and pencil shavings do not.

Q74. What is the aim of the activity on page 3?

Answer: The aim is to show that a magnet has maximum magnetic strength at its poles.

Q75. What is observed when a bar magnet is rolled over iron filings?

Answer: Most of the iron filings stick near the two ends of the magnet.

Q76. What is the aim of the activity on page 4?

Answer: The aim is to show that a freely suspended magnet comes to rest in a particular direction.

Q77. What is observed when a freely suspended magnet is disturbed?

Answer: It turns and comes to rest again approximately in the north–south direction.

Q78. What is the aim of the activity on page 5?

Answer: The aim is to show that like poles repel and unlike poles attract each other.

Q79. What is the conclusion of the activity on page 5?

Answer: North–north and south–south poles repel each other, while north–south poles attract each other.

Q80. What is the aim of the activity on page 8 involving a magnet and a compass?

Answer: The aim is to observe how a nearby magnet affects a compass needle and whether placing certain non-magnetic materials between them changes its deflection.

Q81. What is observed when wood, cardboard, plastic or thin glass is placed between a magnet and a compass?

Answer: In the textbook activity, these materials do not significantly change the compass needle's deflection. They do not block the magnetic field in the way a magnetic shield would.

Part 4: Multiple-choice questions (MCQs)

Test your knowledge

0/15 answered

1. What does a magnet attract?

A. Wood
B. Iron

C. Paper
D. Rubber

2. Which is a natural magnet?

A. Lodestone
B. Plastic ruler
C. Wooden stick
D. Glass sheet

3. Where is a bar magnet strongest?

A. At the centre only
B. At its poles
C. At every point equally
D. Outside the magnet only

4. How many poles does an ordinary bar magnet have?

A. One
B. Two
C. Three
D. Four

5. What happens when two north poles are brought close?

A. They attract
B. They repel
C. They disappear
D. Nothing always happens

6. What happens when a north pole and a south pole are brought close?

A. They attract
B. They repel
C. They lose their poles
D. They become non-magnetic

7. What happens when a magnet is broken into two pieces?

A. Only north poles remain
B. Each piece has north and south poles
C. Only south poles remain
D. Both pieces become non-magnetic

8. Which material is usually non-magnetic?

A. Iron
B. Nickel
C. Cobalt
D. Wood

9. In which direction does a freely suspended magnet settle?

A. East–west approximately
B. North–south approximately
C. Up–down
D. It never settles

10. Which instrument helps us find directions?

A. Thermometer
B. Magnetic compass
C. Measuring tape
D. Barometer

11. What is magnetic induction?

A. Heating a magnet
B. Magnetising a material through the influence of a nearby magnet
C. Breaking a magnet
D. Painting a magnet

12. What is a magnetic keeper?

A. A wooden ruler
B. A piece of soft iron used to help preserve a magnet's strength
C. A plastic cover
D. A compass dial

13. Which object can be used to make a simple floating compass?

A. Magnetised needle on cork
B. Paper clip alone
C. Rubber band
D. Glass rod

14. Which device commonly uses magnets?

A. Speaker
B. Wooden chair
C. Paper notebook
D. Cotton cloth

15. What should we avoid doing to a magnet?

A. Using it carefully
B. Keeping it away from sensitive devices
C. Heating or hammering it strongly
D. Storing it properly
Check answers

Part 5: Fill in the blanks

1. A magnet attracts materials such as iron, nickel and cobalt.
2. A naturally occurring magnet is called a lodestone.
3. The two ends of a magnet are called its poles.
4. A magnet is strongest near its poles.
5. Like poles repel each other.
6. Unlike poles attract each other.
7. A freely suspended magnet comes to rest in the north–south direction.
8. Magnetic poles always exist in pairs.
9. The process of making a material magnetic is called magnetisation.
10. A device used to find directions is called a magnetic compass.
11. A piece of soft iron used to preserve a magnet's strength during storage is called a magnetic keeper.
12. A magnet can help separate iron from scrap materials.
13. Wood and plastic are generally non-magnetic materials.
14. Magnets are used in speakers and electric motors.
15. Strong heating and hammering can weaken a magnet's magnetic properties.

Part 6: True or False

Statement	Answer
1. All metals are attracted by magnets.	False
2. Iron is a magnetic material.	True
3. A bar magnet has north and south poles.	True
4. Like poles attract each other.	False
5. Unlike poles attract each other.	True
6. A freely suspended magnet points approximately north–south.	True
7. Breaking a magnet produces an isolated north pole.	False
8. Magnets are used in electric motors.	True
9. Wood is strongly attracted by an ordinary magnet.	False
10. A magnetic compass contains a magnetised needle.	True
11. Magnets should be heated and hammered for safe storage.	False
12. A magnet can be used to separate iron from mixed waste.	True

Part 7: Quick revision notes

Magnet

Attracts magnetic materials such as iron, nickel and cobalt.

Magnetic poles

A magnet has north and south poles. Its strength is greatest near the poles.

Magnetic force

Like poles repel; unlike poles attract.

Magnetic compass

A magnetised needle aligns approximately north–south.

Care of magnets

Avoid strong heating, hammering and improper storage. Keep magnets away from sensitive electronic devices.

Exam tip: Learn the definitions of magnetism, magnetic materials, magnetic poles, magnetic induction, magnetic compass and magnetic keeper. Practise explaining the two activities showing attraction and repulsion, and remember the five common uses of magnets.

These notes cover the questions visible in the uploaded chapter and provide additional revision questions from all the major topics.